Forest Group — Discussion Topics by Project Phase

> **"The forest our ancestors planted — powering the village they built."** 先人が植えた森が、今、村に豊かさをもたらす。

This document lists the topics the forest working group needs to address, mapped to the four project phases (Apr 2026 – Sep 2028).

Key Partners & Contacts

Organisation	Role	Contact status	Notes
御杖村森林組合 (mitsue-kanko)	Primary thinning & chip production partner	✅ Initial contact done (2026-06-22)	Clarify role: partner, contractor, or hybrid
More Trees / Carbontribe — Tenkawa	Native reforestation model + J-Credit	❌ Not yet contacted	Active project next village over; Ryuichi Sakamoto NGO; may let us join their J-Credit registration
Niwamori.org — Nara	Miyawaki reforestation expertise + food forest	❌ Not yet contacted	Contact: Jérôme Floerke; uses Miyawaki method (native canopy in 20–30 yrs); active in Nara
奈良県森林環境課	Prefectural grants + 林野庁 subsidy gateway	❌ Not yet contacted	Administers national broadleaf promotion subsidies locally
林野庁 (Forestry Agency)	National thinning subsidies + broadleaf programme	❌ Not yet contacted	Dedicated 広葉樹 programme; apply via prefecture

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Phase 0 — Pre-Foundation *(M1–M3, Apr–Jun 2026 — now)*

- ~~Identify the forestry co-op contact and open the conversation~~ ✓ *Initial contact with mitsue-kanko done (2026-06-22)*
- Contact More Trees / Carbontribe (Tenkawa) — explore J-Credit partnership
- Contact Niwamori.org — explore Miyawaki replanting partnership for native species
- Contact 奈良県森林環境課 — register project interest, ask about 広葉樹 subsidies
- Get a rough landowner map (who owns which parcels)
- Agree that the new NGO (Tier 1, 一般社団法人) will be the contracting party before anything is signed

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Phase 1 — Foundation *(M4–M9, Jul–Dec 2026 · ¥1.5M forestry feasibility + reforestation planning)*

> ⚠ **Budget note:** The ¥1.5M feasibility budget must cover reforestation planning and coordination costs in addition to chip production feasibility. Native species replanting (Miyawaki method) requires specialist input

> for site assessment, species selection, and deer-fencing design. This line item should be reviewed upward — recommend ¥2.0–2.5M.

Supply Chain: Chip Production

- Who does the thinning — mitsue-kanko, private contractors, or hybrid?
- Equipment needed: feller, forwarder, chipper (own vs. rent vs. contract)
- Chip sizing standard required by the CHP (P31S / P45, ~30–50 mm)
- Annual volume target — how many ha/year to feed 50 kWel continuously (~40–50 kg/hr)?
- Benchmark from Mishima Town study: 700–800 tonnes/year for ≤50 kW CHP; chip cost target ≤¥7,000/m³
- Transport route from forest to processing/storage site

Drying & Storage

- Green chips arrive at ~40–50% moisture; CHP requires ≤15% — who/what dries them?
- Drying method: passive (shed + airflow), active (waste heat from CHP loop), or kiln
- Storage volume: how many weeks of buffer needed?
- Location: co-located with CHP plant or at a satellite forest site?
- Fire safety / permitting requirements for chip storage shed

Native Species Reforestation Plan

This is a multi-generational commitment, not a Phase 3 task. Planning must start in Phase 1.

Who does the replanting:

- mitsue-kanko for physical planting labour (to confirm)
- Niwamori.org as Miyawaki method specialist / coordinator (to confirm via contact)
- Village volunteer planting days as community engagement (optional supplement)

Method — Miyawaki:

- Dense planting of diverse native broadleaf species (30–50 seedlings/m²)
- Felled sugi trunks repurposed as raised beds along contours — erosion control + soil building
- Branches composted back into soil
- Self-sustaining native canopy achieved in **20–30 years**
- First visible results (canopy closure, biodiversity return) within **3–5 years**
- Source / precedent: [Niwamori.org Nara]

Timeline per parcel:

- Year 0: Thinning complete → immediate Miyawaki planting on cleared ground
- Years 1–5: Annual weeding + deer-fencing maintenance (labour cost; must be budgeted)
- Years 5–10: Canopy closing; maintenance reducing
- Years 20–30: Self-sustaining native forest
- 2050: Full ecological restoration (aligns with founding charter vision)

Species selection:

- To be defined with Niwamori.org in Phase 1
- Must include natural deer/bear food sources (acorn-bearing species: コナラ, クヌギ) to reduce wildlife crop raids — explicitly stated in founding charter

What needs to be costed (currently missing from budget):

- Native broadleaf seedlings: ~¥300–800/seedling vs. ¥100–200 for sugi
- Deer + boar fencing per parcel (Mitsue has documented wildlife pressure)
- Annual maintenance labour: years 1–5 per planted parcel
- Miyawaki specialist coordination fee (Niwamori.org or equivalent)
- Long-term monitoring and J-Credit reporting
- Which parcels are prioritised for thinning? (mapping of degraded sugi stands)
- Thinning schedule — years 1–10 rotation plan
- Who monitors canopy recovery and certifies progress?
- Link to J-Credit methodology: baseline, monitoring, reporting intervals

J-Credit & Carbon Revenue

- Explore joining More Trees / Carbontribe Tenkawa registration rather than starting new
- If new registration: which methodology? (森林経営活動 / ARR?)
- Who manages annual monitoring reports?
- Credit ownership: village, NGO (Tier 1), or shared with landowners?
- Expected tCO_2/yr and revenue estimate (current J-Credit price ~¥3,000–6,000/t)

Land Access & Contracts

- Inventory of landowners whose parcels are involved
- Harvest rights agreement: terms, duration, compensation model (per m^3 or % of chip revenue)
- Right-of-way for forest roads and transport
- Confirm NGO as the contracting party before signing
- Any village-owned forest land usable without private contracts?

Forestry Co-op Relationship (御杖村森林組合 / mitsue-kanko)

- Is mitsue-kanko a partner, contractor, or competitor?
- Co-op capacity: do they have chippers and operators available?
- Do they have experience with native species replanting, or only sugi?
- Profit-sharing or service-fee model?

Economics & Feasibility *(Gate 2 decision — Dec 2026)*

- Cost per tonne of dried chip delivered to CHP gate
- Independent feasibility check: does internal fuel cost close the energy margin?
- Full reforestation cost model: seedlings + fencing + maintenance years 1–5 + coordination
- Subsidy options: 林野庁 thinning + 広葉樹 subsidies, MoE 交付金 (2/3–3/4), J-Credit stacking
- **Kill criterion:** if feasibility shows negative base-case surplus → descope to forestry+solar only

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Phase 2 — Pilot Design *(M10–M18, Jan–Sep 2027 · ¥3.5M for permitting incl. forestry)*

- Obtain forestry practice permits (伐採届 — village or prefecture?)
- Finalise chip storage shed design and fire-safety approval

- Choose CHP unit based on feasibility results (Spanner Re², Volter, Fröling, etc.)
- Submit J-Credit registration paperwork (or join More Trees existing registration)
- Secure 林野庁 thinning subsidies + 広葉樹 programme grants + MoE 交付金
- Finalise Miyawaki replanting plan with Niwamori.org: species list, fencing design, per-parcel schedule
- Procure native seedlings (long lead time — order in Phase 2 for Phase 3 planting)

Phase 3 — Pilot Build *(M19–M30, Oct 2027–Sep 2028)*

> ⚠ **Budget note:** WBS 5.6 currently allocates ¥25M for "forestry operations (5–10 ha harvest + replant)." This must be revised to separately cover: (a) thinning & chip production, (b) native Miyawaki replanting, (c) deer/boar fencing, (d) years 1–5 maintenance provision. Recommend splitting into two line items and flagging for Rev 2 (M9 Dec 2026).

- **Actual thinning begins** (5–10 ha, M19–M23 to avoid typhoon season)
- Chip production, drying infrastructure, and storage shed constructed
- CHP commissioned and connected to data center baseload
- **Miyawaki replanting** executed parcel-by-parcel immediately after each thinning pass
- Deer/boar fencing installed on all replanted parcels
- J-Credit monitoring cycle starts
- Year 1 native seedling maintenance programme begins

Ongoing (Phase 4+, from 2028)

- Annual maintenance of Miyawaki plantings (years 1–5 per parcel: weeding, fencing checks)
- J-Credit annual monitoring and reporting
- Canopy progress photography + biodiversity recording (community engagement + grant reporting)
- Second thinning rotation planning (next 5–10 ha)
- 2050 target: full native forest restoration across all thinned parcels

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Roles & Immediate Next Steps

Action	Owner	By when
Name a forest working group lead	Board / founding team	Phase 0
Contact More Trees / Carbontribe Tenkawa	Working group lead	Jun–Jul 2026
Contact Niwamori.org (Jérôme Floerke)	Working group lead	Jun–Jul 2026
Contact 奈良県森林環境課	Rep. Director	Jul 2026
Produce forest parcel map + landowner contact list	Working group lead	Phase 0 end
Confirm NGO as contracting party	Legal / NGO setup team	Before any contract
Procure forestry feasibility study (incl. reforestation plan)	Rep. Director	Q3 2026 (M4)
Review + uplift WBS 5.6 budget for native replanting costs	Rep. Director + EVM	M9 (Dec 2026, Rev 2)

Action	Owner	By when
First thinning contract signed	NGO (Tier 1)	Phase 1 end (M9)
Order native seedlings	Niwamori.org / co-op	Phase 2 (long lead time)

> **Critical path note:** The Phase 1 forestry feasibility study must cover both chip production economics AND a full native reforestation cost model. WBS 5.6 (¥25M) is likely underestimated once native seedlings, deer fencing, and 5-year maintenance are included. Gate 2 (Dec 2026) is the moment to correct this before Phase 3 spend is committed.

Sources & Precedents

- More Trees / Carbontribe — Tenkawa Village, Nara — active native reforestation + J-Credit in the same mountain range
- Niwamori.org — Miyawaki Food Forest, Nara — Miyawaki method specialist active in Nara; 20–30 yr native canopy timeline
- Niwamori Sustainable Forest Proposal (Nov 2024)
- 林野庁 広葉樹の取組 — national broadleaf forest promotion subsidies
- 奈良県森林環境課 — prefectural forestry grants gateway
- Ooba, Nakamura & Togawa (2020), *Promoting Local Revitalization to Solve Issues on Degraded Forests in Japan* — Mishima Town CHP precedent; 700–800 t/yr biomass, ≤¥7,000/m³ chip cost benchmark